

- Fenwick Tree supported $O(\log n)$ point updates and prefix/range sum queries.
- Total score (all 8 players): 980
- Score for players 3–6: 525
- After player 4 gains 50 pts: total becomes 1030

Time Complexity:

- Update: $O(\log n)$ | Prefix Sum Query: $O(\log n)$ | Range Query: $O(\log n)$

Compared to a simple array ($O(n)$ for range sum), Fenwick Tree performs significantly better under high-frequency MMO server traffic where real-time leaderboard accuracy is critical.

GitHub Link:

https://github.com/egotpat-gif/DSAIL-Case-Studies/tree/main/DSA_CO2_FENWICKTREE_CASESTUDY

Student Signature	Faculty Signature